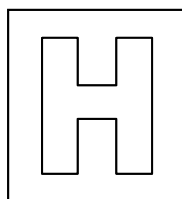


Candidate Name: _____

Class Adm No

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2013 Preliminary Examination 2 Pre-University 3

H2 Biology

9648/02

17 September 2013

2 hours

Additional Materials: Writing paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Index number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question. At the end of the examination, fasten all your work securely together.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
Section B	
Total	

This question paper consists of 24 printed pages.

[Turn over

Answer **all** questions in this section.

- Table 1.1 shows how the foxes were put into categories according to their tameness.

tameness class	description of behaviour towards humans
3	Not tame - these foxes run away from humans or bite when handled.
2	Neutral - these foxes allow handling by humans but show no emotionally friendly response.
1	Tame - these foxes are friendly to humans. They wag their tails and whine for attention.
elite	Very tame - these foxes are eager for human elite contact. They whimper to attract attention and sniff and lick humans.

(a)

- | | | |
|-------|---|-----|
| | artificial selection / selective breeding ; | [1] |
|-------|---|-----|

- idea that females produce only a few offspring ;
(so) more females (than males) needed to maintain numbers (each generation) ;
(20% females chosen as) inbreeding / genetic problems, if breeding population is too small;
(5% males chosen as) selection pressure stronger if fewer (tamest) are used;

- (b) Table 1.2 shows the number of foxes in the elite tameness class during the long-term experiment.

Table. 1.2

Number of generations	Foxes in elite class (%)
10	18
20	35
35	75

Discuss what the results shown in Table 1.2 suggest about the causes of variation in tameness behaviour in silver foxes.

- 1 (mostly) genetic ;
 2 as can be selected for / selective breeding increases frequency ;
 3 allele(s) for tameness ;
 4 (from) mutation ;
 5 query role of environment / learning ;
 6 ref. DRD4 / dopamine receptor ;
 **R** environment also given as cause
 **R** genetic drift
 [3]

- (c) As tameness increased in the silver fox population over the years, it was noticed that other phenotypic traits also became more common.

Table 1.3 compares the frequency of these traits in a control group of silver foxes that had not been used in this long-term breeding experiment and in the tame population of foxes.

Table. 1.3

Phenotypic trait	Animals showing trait (per 100 000)		Percentage increase in trait
	Control Population	Tame Population	
White patch of fur on head	710	12 400	1 646
Floppy ears	170	230	35
Short tail	2	140	6 900
Curly tail	830	9 400	1 033

Students were asked to suggest a variety of genetic hypotheses to explain why these traits become more common in tame foxes. Their suggestions were:

Linkage Epistasis Inbreeding Genetic Drift

Select one hypothesis from the list and explain how it could account for the data in Table 1.3.

linkage

tameness genes and genes for these traits on same chromosome ;
(so) inherited together ;

epistasis

(product of) one gene affects expression of another ;
via enzyme pathway ;

inbreeding

(hidden / masked) recessive alleles ;
selected for, as well / unintentionally ;
more chance homozygous as, small gene pool / parents related ;

ACCEPT idea of (recessive) allele inherited from both parents because, they are closely-related / small gene pool / reduced genetic diversity

genetic drift

random / chance (which alleles, present / passed on) ;
(effect stronger because) small breeding population ;

.....
.....
.....
.....
..... [2]

- (d) Similar changes in tameness, colour and body shape are believed to have occurred in the 11 000 year period during which the grey wolf species, *Canis lupus*, evolved into the domesticated dog species, *Canis familiaris*.

Suggest how different types of isolating mechanism allowed dogs to evolve separately to wolves.

- | | | |
|-------|---|----------|
| | 1 geographic ; | |
| | 2 wolves avoid human settlements / dogs confined by humans ; | |
| | 3 behavioural ; | |
| | 4 detail / description ; eg differences in, pheromones / courtship | |
| | 5 mechanical ; | |
| | 6 idea of different size of wolves and some small dogs ; ACCEPT different genitalia | |
| | 7 gamete incompatibility ; | |
| | 8 possibility of different chromosome numbers ; | |
| | 9 seasonal / temporal ; | |
| | 10 different breeding, seasons / times ; idea that dogs breed all year round / wolves breed once a year | [3] |

- (e) Interbreeding between members of the wolf species and some dogs has been reported.

However, there are some large breeds of dogs that cannot breed successfully with small dog breeds.

Use this information and your own knowledge to explain the problems of classifying wolves and different dog breeds according to:

- the biological species concept

and

- the phylogenetic species concept.

- | | | |
|-------|---|-------|
| | biological species concept | .. |
| | 1 (members of same species) need can interbreed to produce fertile offspring ; | .. |
| | 2 not all dog breeds can do this therefore not same species ; | .. |
| | 3 dog and wolf can so they should be same species ; | .. |
| | phylogenetic species concept | .. |
| | 4 idea that dogs and wolves monophyletic group / tip of phylogeny ; | .. |
| | 5 genetic differences, between dogs and wolves small ; | .. |
| | 6 gene flow between wolves → big dogs → little dogs / analogous to ring species ; | .. |
| | 7 (PSC) one species (with a lot of phenotypic variation) ; | .. |
| | | .. |
| | | |
| | | [4] |

[Total: 15]

2.

- (a) The brain depends on a constant supply of oxygen for aerobic respiration. Anaerobic respiration is not sufficient to keep neurones in the brain alive. This is because neurones require especially large amounts of ATP. Up to 80% of the ATP is used to provide energy for the Na^+ / K^+ pump.

When a person suffers a stroke, blood flow to part of the brain is stopped, so some neurones receive no oxygen. ATP production by oxidative phosphorylation stops. Fig. 2.1 shows some of the ways in which the lack of ATP affects a neurone in the brain.

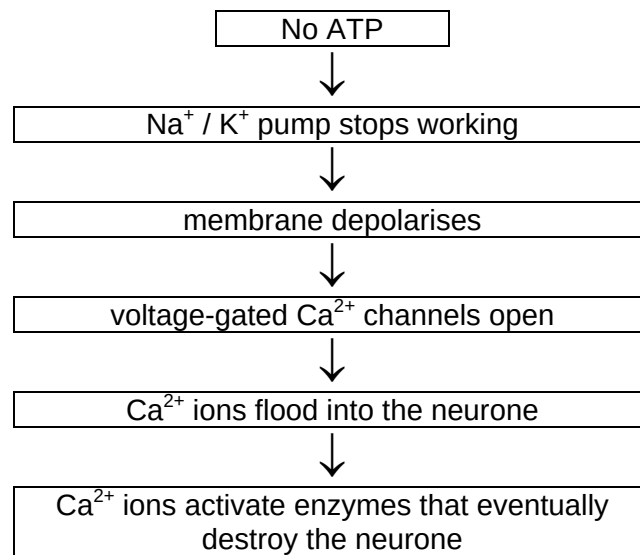


Fig. 2.1

- (i) Explain why the membrane of the neurone depolarises when the Na^+ / K^+ pump stops working.

- | | | |
|-------|---|-----|
| | 1. (when pump stops working), resting potential not maintained | ... |
| | or | |
| | pump usually maintains the resting potential; | ... |
| | 2. (during resting potential) membrane polarised | ... |
| | or | |
| | positive charge outside (neurone) / negative charge inside (neurone) / -70mV | ... |
| | inside neurone relative to outside / potential difference across membrane; | |
| | 3. (when pump stops working), ions (only) move by diffusion; | ... |
| | 4. Na^+ into the neurone; | ... |
| | 5. outward diffusion of K^+ is limited / K^+ stay in neurone; | ... |
| | 6. ref. non voltage-gated channels; | ... |
| | 7. (eventually) inside of the neurone, becomes less negative / contains | |
| | (relatively) more positive ions | [4] |
| | or | |
| | there is a reduced potential difference across the membrane; | |

- (ii) Suggest why calcium ions flood into the neurone when the Na^+ / K^+ pump stops working.

- | | | |
|-------|--|-----|
| | 1. voltage gated (calcium) channels open; | ... |
| | 2. (calcium ions move in) by diffusion / move down their concentration gradient; | ... |
| | | ... |
- [2]

- (b) The freshwater turtle, *Trachemys scripta*, is able to survive for long periods in conditions of very low oxygen concentration. As in humans, the rate of activity of the Na^+ / K^+ pump in the neurones in its brain falls sharply. However, in turtles this does not result in damage to these cells.

A better understanding of how the neurones in the turtle's brain survive in these conditions could lead to new treatments for people who have suffered a stroke.

Experiments show that, in turtle brain neurones, in conditions of low oxygen availability:

- most ion channels in the cell surface membranes immediately close
- after about four hours, the quantity of mRNA involved in the synthesis of proteins used to build ion channels, falls to less than one fifth of normal concentrations.

- (i) Suggest how the closure of ion channels in the neurones of the turtle in very low oxygen concentrations could allow the cells to survive.

- | | |
|---|---|
| 1. Na^+ / K^+ , cannot move through membrane; | . |
| 2. so potential across membrane maintained even when pump stops / so membrane depolarisation does not happen; | . |
| 3. calcium ions cannot enter cell; | . |
| 4. so, (destructive) enzymes not activated; | . |
- [2]

- (ii) Suggest what causes the quantity of mRNA for protein channels to fall.

- | | | |
|-------|--|-----|
| | 1. gene (for protein channels), expressed less / switched off; | ... |
| | 2. transcription, reduced / stopped; | ... |
| | 3. AVP; e.g. reduced aerobic respiration / less ATP, for transcription | ... |
- [2]

- (c) Some synapses in the brain use the neurotransmitter dopamine. After the postsynaptic membrane has been depolarised, dopamine leaves the receptor proteins and moves back into the presynaptic neurone through specific transporter proteins.

Schizophrenia is a condition in which there is a higher than usual concentration of dopamine in certain areas of the brain. The drug phenothiazine has a similar shape to dopamine and is used to treat schizophrenia.

Suggest and explain what occurs at the synapse when phenothiazine is used in the treatment of schizophrenia.

.....	binds to/blocks, dopamine receptors (on postsynaptic membrane);	
.....	prevents depolarization (of postsynaptic membrane);	
.....	reduces effect of dopamine;	
.....	R reduces amount of dopamine	

..... [2]

- (d) DRD4 is a dopamine receptor in humans. The gene coding for the DRD4 receptor has a large number of alleles, of which an individual can have only two.

Three alleles of the DRD4 receptor gene have the following mutations:

- a single base substitution
- a 21 base-pair deletion
- a 13 base-pair deletion.

The 13 base-pair deletion has the most serious consequences for the structure of the DRD4 receptor protein. Suggest why this is so.

...	<i>ref 13 base deletion</i>	
...	<u>frame shift</u> /alters <u>reading frame</u> (after mutation);	
...	(so) all amino acids different after mutation;	
...	3-D shape/tertiary structure, of protein changed;	
...	(whereas) 21 base-pair deletion, loses 7 amino acids/no frame shift;	
...	(whereas) substitution, may change only one amino acid/may be silent;	
...		
...		

..... [3]

- (e) One allele of the *DRD4* gene has been found more frequently amongst individuals whose personality is described as 'impulsive and exploratory'.

Describe the mechanism whereby an allele such as this could have become common in the human population.

.....

...	increased chances of, survival/breeding/mating;	
...	provides a selective advantage;	
...	allele passed on (to next generation);	
...	allele increases in frequency over time;	
...	natural selection;	

.....

..... [3]

[Total: 18]

3. Bone marrow contains stem cells that divide by mitosis to form blood cells. Each time a stem cell divides it forms a replacement stem cell and a cell that develops into a blood cell.

Fig. 3.1 shows changes in the mass of DNA in a human stem cell from the bone marrow during three cell cycles.

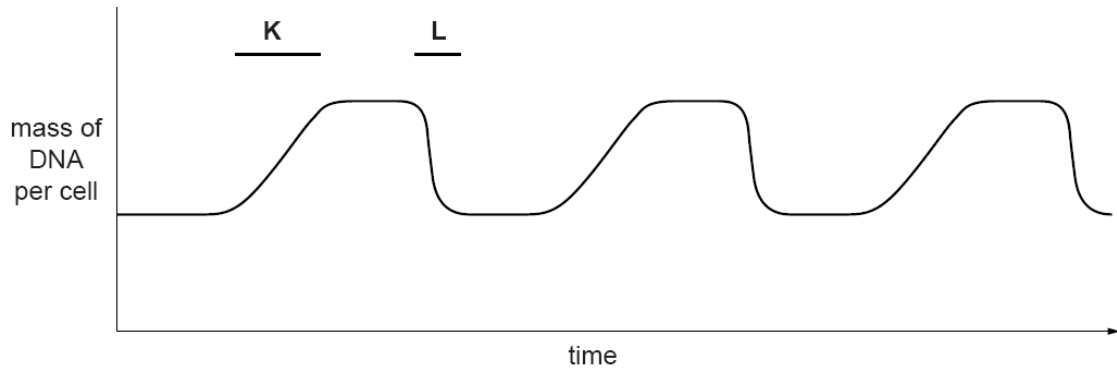


Fig. 3.1

- (a) With reference to Fig. 3.1, state:

- (i) what happens to bring about the changes in the mass of DNA per cell at **K** and at **L**.

K	K – (DNA) replication / synthesis / described ;	
.....	L – cytokinesis / cytoplasmic division / cell division ;	
L		
.....		... [2]

- (ii) how many blood cells are formed from the stem cell in the time shown.

.....	3 ;	 [1]
-------	-----	-----------

- (iii) what happens to the number of chromosomes in the stem cell.

remain the same / stays constant / stay at 46 / AW ; ignore description of events occurring before and during mitosis	[1]
--	-----

Stem cells in bone marrow give rise to phagocytes, B-lymphocytes and T-lymphocytes.

(b) During an immune response, cells divide by mitosis.

Suggest how mitosis is involved in an immune response.

- | | |
|--|--|
| .. transcription (of specific genes) ; A reference to gene switching
.. protein / polypeptide, synthesis ; A translation
.. lymphocytes / B -cells / T-cells / divide (by mitosis) ;
.. producing, clone / many cells ;
..
..
.. |
.....
.....
.....
.....
..... |
|--|--|
- [3]

A protein, mitosis-promoting factor (MPF), has been identified in cells. MPF is a globular protein made from two polypeptide chains.

The presence of MPF is known to cause prophase to start.

(c) Describe the changes that occur during prophase in an animal cell.

- | | |
|--|---|
|
appearance of chromosomes / condensation of chromatin / AW ;
chromosomes visible as two, sister chromatids / chromatids joined by a centromere ;
spindle formation / spindle fibres form / microtubules assemble / AW ;
centrioles, move to / reach, opposite poles ;
R sides / ends
disappearance of nucleolus ;
disassembly / breakdown of, nuclear envelope ;
A nuclear membrane
..... |
.....
.....
.....
.....
.....
..... |
|--|---|
- [4]

- (d) MPF normally begins to break down and stops functioning during anaphase.

Suggest the possible consequences of MPF not breaking down.

mitosis / prophase, will begin again, too soon / immediately ;
uncontrolled / repeated, cell division / mitosis ;
ignore (risk of), tumour formation / cancerous growth
ref. to consequences on the timing of the cell cycle ;

..
..
..
..
..

..... [3]

[Total: 14]

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4. A type of pheasant occurs in a range of colours, especially when bred in captivity. It may, for example, have green or purple plumage.

Sometimes when a green male is crossed with a green female all the offspring, male and female, are green. However, sometimes a green male crossed with a green female results in offspring in which the majority of the offspring are green, but in which some of the females are purple, as shown in Table 4.1.

Table. 4.1

Phenotype	Number of offspring
green male	7
green female	3
purple female	4

(a)

- (i) Using the results shown in Table 4.1, a geneticist proposed a hypothesis which predicted a 3:1 ratio of green:purple phenotypes. She used the χ^2 test to test her hypothesis

Complete Table 4.2 to calculate χ^2 .

Table. 4.2

	Green Plumage	Purple Plumage
Observed (O)	7 + 3 = 10	4
Expected (E)		
$O - E$		
$(O - E)^2$		
$\frac{(O - E)^2}{E}$		
$\Sigma \frac{(O - E)^2}{E} = \chi^2$		

[2]

- (ii) Draw a conclusion from your calculation in Table 4.2, given that the critical value for χ^2 at $p = 0.05$ with one degree of freedom is 3.84. Explain why this conclusion should be treated with caution.

.....
 [3]

- (b) Plumage colour in pheasants is sex-linked.

In birds, the sex chromosomes are referred to as W and Z, rather than Y and X as in mammals. The W chromosome has no genes that affect plumage colour. The heterogametic sex is the female, not the male. Thus the male has two Z chromosomes (ZZ) and the female has one W and one Z chromosome (WZ).

- (i) Use a genetic diagram to explain the results in Table 4.1.

parental phenotypes: green male × green female

parental genotypes: $Z^G Z^g$ × WZ^G

gametes: Z^G Z^g × W Z^G

Punnett square:

		green male	
		Z^G	Z^g
green female	W	WZ^G	WZ^g
	Z^G	$Z^G Z^G$	$Z^G Z^g$

green male = $Z^G Z^G$ and $Z^G Z^g$

green female = WZ^G

purple female = WZ^g

2 green male: 1 green female: 1 purple female

- choice of appropriate symbols for the gene for plumage and its alleles ;
e.g. G/g
accept ecf in later parts of question if inappropriate symbols used, e.g. G and P
- gametes of both parents correct and offspring genotypes derived correctly ;
accept X and Y instead of W and Z if used correctly, e.g. male = XX,
female = XY
- correct link between genotype and phenotype indicated ;

[3]

- (ii) Using the same symbols as in (b)(i), indicate the genotypes and phenotypes of the parents which could give rise to purple male offspring.

$Z^G Z^g \times Z^g W$ green (heterozygous) male $\times$ purple female ;

$Z^g Z^g \times Z^g W$ purple male $\times$ purple female ;

accept ecf from (b)(i) for symbols of alleles

accept $Z^G Z^g$ green (heterozygous) male, $Z^g Z^g$ purple male, $Z^g W$ purple female
if not written out as crosses

[2]

[Total: 10]

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5.

- (a) An experiment was carried out to investigate the effect of temperature on the rate of oxygen consumption of the lizard, *Sauromalus hispidus*. The body temperature of a lizard varies with environmental temperature.

Several lizards were fitted with small, airtight masks that covered their heads. Air was supplied inside the mask through one tube, and collected through another. The differences between oxygen concentrations in the air supplied for inhalation and the exhaled air enabled the researchers to measure the rate of oxygen consumption of the lizards.

The rate of oxygen consumption of each lizard was measured when it was at rest and when it was running. Measurements were made at different temperatures ranging from 15 °C to 40 °C.

Fig. 5.1 shows the results.

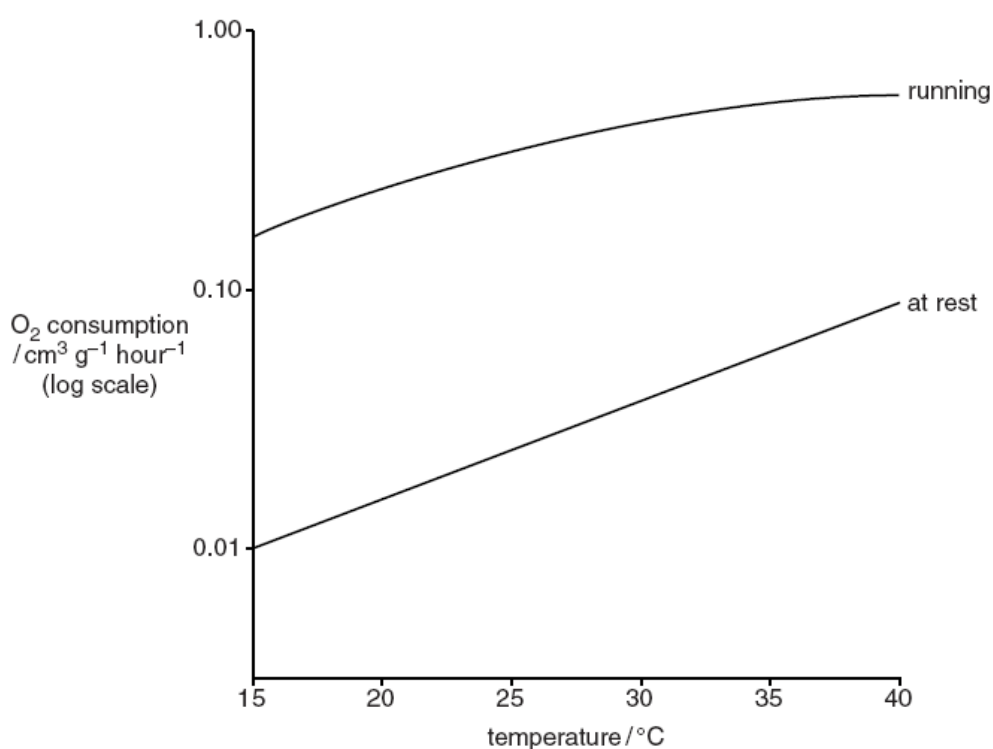


Fig. 5.1

- (i) Running requires rapid use of ATP by muscle cells in the legs and heart of a lizard.

With reference to the events occurring inside a mitochondrion, explain why a faster use of ATP requires a greater rate of oxygen consumption.

- | | | |
|-------|--|-------|
| | 1. ATP is made, in the electron transport chain/by oxidative phosphorylation; | |
| | 2. oxygen is the final electron acceptor; | |
| | 3. in the, inner membrane of the mitochondrion/cristae; | |
| | 4. transfer of electron (between electron carriers) provides energy; | |
| | 5. energy used to pump hydrogen ions (into intermembrane space); | |
| | 6. creates proton gradient; | |
| | 7. diffusion of hydrogen ions down their electrochemical gradient causes ATP to be synthesised; | |
| | 8. ref. chemiosmosis/ATP synthase/stalked particles; | |
| | 9. idea that if less oxygen (consumed/available) then fewer electrons transferred along the chain; | |
| | | |

..... [4]

- (ii) Explain the effect of temperature on the rate of oxygen consumption in *Sauromalus* when at rest.

- | | | |
|-------|--|-----|
| | 1. at high temperatures, reactions/enzyme activity/metabolism, faster; | ... |
| | 2. because, molecules/enzymes/substrates, have more kinetic energy; | ... |
| | 3. more frequent collisions; | ... |
| | 4. therefore, respiration/Krebs cycle/electron transport chain/production of reduced NAD, take place at a faster rate; | ... |
| | 5. idea of increase in rate of anabolic reactions (requiring more ATP); | ... |
| | | ... |

..... [3]

- (b) In the light-independent stage of photosynthesis, the enzyme rubisco catalyses the combination of carbon dioxide with ribulose biphosphate, RuBP. When the carbon dioxide concentration within the leaf is very low, rubisco tends to combine oxygen, rather than carbon dioxide, with RuBP. This process is called photorespiration. It reduces carbon dioxide assimilation and therefore reduces crop yields.

Photorespiration is most likely to happen in hot, dry conditions.

Suggest why photorespiration is most likely to take place in hot, dry conditions

.....	1. stomata closed ;	..
.....	2. to reduce transpiration / to avoid too much loss of water ;	..
.....	3. so carbon dioxide cannot enter the leaf ;	..
.....	4. so carbon dioxide concentration (in leaf / in chloroplast) becomes very low ;	..
.....		..
.....		..
.....		..

.....

.....

..... [3]

- (c) It is expected that the carbon dioxide concentration in the atmosphere will increase in the future, which would be expected to increase rates of photosynthesis in many crop plants.

Investigations were carried out into the effect of increased carbon dioxide concentration on the rate of photosynthesis in maize.

- Maize plants were grown in open-air trials, in the same field and were exposed to the same changes in the weather.
- 50% of the plants were exposed to a normal carbon dioxide concentration.
- 50% of the plants were exposed to an increased carbon dioxide concentration.
- The rate of photosynthesis was measured as the net assimilation rate of carbon dioxide.
- Measurements were made at three-hourly intervals between 0700 hours and 1900 hours on three different days.

The results are shown in Fig. 5.2.

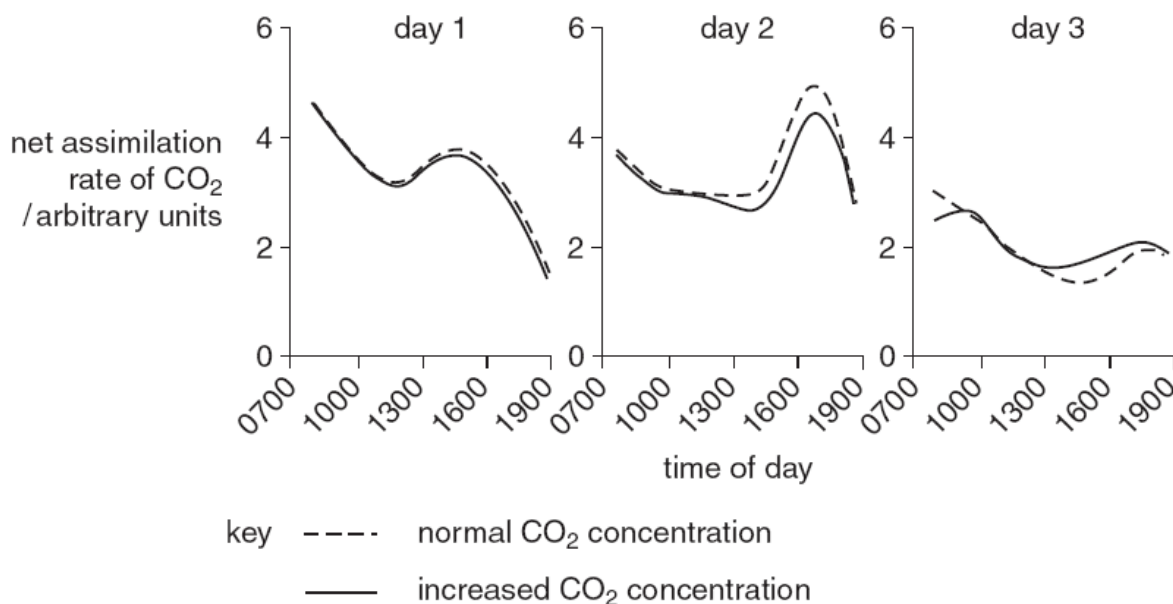


Fig. 5.2

- (i) Suggest an explanation for the lack of effect of carbon dioxide concentration on the rate of photosynthesis in maize plants, shown by these results.

..... 1. CO₂ concentration is always high ;
 2. CO₂ not limiting ;
 3. another factor / light intensity / temperature, limiting ;
 4. no photorespiration ;

[2]

- (ii) Suggest one explanation for the changes in the rate of photosynthesis between 0700 hours and 1900 hours on day 1.

..... 1. idea of change in temperature ;
 2. affects, light independent / light dependent, stage (of photosynthesis) ;
 or
 3. idea of change in light intensity ;
 4. affects light dependent stage (of photosynthesis) ;

[2]

[Total: 14]

6. A study finding published in the journal PLoS Genetics in 2013 was the first to show exercise alters DNA methylation related to fat storage.

The finding adds to a growing body of knowledge about epigenetics that show though we cannot change our DNA we can change the way our genes are expressed to fight disease.

- (a) Describe and explain the effect of DNA methylation in eukaryotic genome.

.....	DNA methylation leads to a more compact chromatin structure / chromatin with a closed conformation / heterochromatin;	
.....	Addition of methyl group to certain bases (usually cytosine) of DNA;	
.....	DNA methyltransferase catalyze the transfer of a methyl group to DNA;	
.....	Transcription factors and RNA polymerase unable to bind to DNA;	
.....	Methylated DNA attracts other proteins which and in turn recruits histone deacetylation enzymes;	
.....		.. [3]

- (b) Suggest how DNA methylation enables cells to differentiate.

.....	Unrequired genes are silenced/not expressed/switched off when methylated;	
.....		.. [1]

In the 1950s, Erwin Chargaff determined the relative quantities of the four bases in DNA in different organisms. His results provided important evidence for the model of DNA proposed by James Watson and Francis Crick in 1953. Some of Chargaff's data is shown in Table 6.1.

Table. 6.1

organism	percentage of adenine	percentage of thymine	percentage of guanine	percentage of cytosine
<i>Escherichia coli</i> (bacterium)	24.7	23.6	26.0	25.7
a yeast	31.3	32.9	18.7	17.1
wheat	27.3	27.1	22.7	22.8
octopus	33.2	31.6	17.6	17.6
sea urchin	32.8	32.1	17.7	17.3
chicken	28.0	28.4	22.0	21.6
human	29.3	30.0	20.7	20.0

- (c) Explain how the data in Table 6.1 helps to confirm the arrangement of bases in DNA.

.....	1 (named) base / nucleotide, pairing ;	
.....	2 purine – pyrimidine ;	
.....	3 percentage of A = percentage of T ; A very similar	
.....	4 percentage of C = percentage of G ; A very similar	
.....	5 data quote in support ;	

.....

..... [3]

- (d) Table 6.2 shows Chargaff's data for a virus.

Table. 6.2

organism	percentage of adenine	percentage of thymine	percentage of guanine	percentage of cytosine
a virus	24.0	31.2	23.3	21.5

- (i) State how the result for the virus differs from the results for all the organisms given in Table 6.1.

..	idea that	
..	percentages of, A and T / C and G, are not the same / three percentages are	
..	similar;	
..		

..... [1]

- (ii) Suggest why the results for the virus are different from all the other organisms.

.....	single-stranded DNA / not double-stranded / not a double helix ;	
.....	A may be other bases ;	
.....		

..... [1]

[Total: 9]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

7.

(a) Outline the molecular structure of phospholipids in relation to their function in cell membranes. [7]

(b) Explain the role of the Golgi body and its link to the rough endoplasmic reticulum. [7]

(c) Describe the structure of ribosomes. [6]

[Total: 20]

8.

(a) Explain the effects of competitive and non-competitive inhibitors on the rate of enzymatic activity. [6]

(b) Explain the mode of action of enzymes in terms of specificity and activation energy. [10]

(c) Describe the effect of temperature on enzyme-catalyzed reactions. [4]

[Total: 20]

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9648/02/0/N/12 Qns 8a

Outline the molecular structure of phospholipids in relation to their function in cell membranes. [6]

Structure

- i) 2, fatty acid / hydrocarbon, chain / tails ;
- ii) a phosphate group ;
- iii) (most) contain, nitrogen / choline (attached to phosphate in, head / polar portion) ;

Function

- can form a bilayer ;
- link between, hydrophobic core / AW, and barrier to water-soluble substances ; A polar / ionic
- idea of, hydrophilic / phosphate, head, forming H bonds with water ;
- A facing, water / watery environment / aqueous environment / cytoplasm / cytosol ref. contribution to fluid nature of membrane ;
- further detail ; e.g. mainly saturated fatty acids, less fluid e.g. mainly unsaturated fatty acids, more fluid
- ref. to control over membrane protein orientation ; e.g. hydrophobic –

9648/02/0/N/12 Qns 8b

Explain the role of the Golgi body and its link to the rough endoplasmic reticulum. [7]

Role

1. processing/modification/AW / packaging, of, proteins/ molecules ;
2. description of modification e.g. glycosylation
3. production of, secretory/ Golgi, vesicles
4. production of lysosomes

Link to rER

- (i) Proteins from the ER enter at its cis face (entry face), which is convex and usually oriented toward the nucleus.
- (ii) They are then transported through the Golgi and exit from its concave trans face (exit face).
- (iii) protein synthesized in Rough ER
- (iv) creates transport vesicle by pinching off a portion of its own membrane.
- (v) vesicle then moves either to the cell surface membrane or to the Golgi Apparatus.

9648/02/0/N/12 Qns 8c

Describe the structure of ribosomes. [6]

- (i) 2 separate subunits - Large unit and small unit
- (ii) subunits bind together during protein synthesis
- (iii) made up of ribosomal RNA (rRNA) and protein
- (iv) Free ribosomes suspended in the cytosol
- (v) bound (fixed or attached) ribosomes attached to the outside of the endoplasmic reticulum or nuclear envelope
- (vi) two types of ribosomes that differ in size and mass: 70S (found in prokaryotic cells) and 80S (found in eukaryotic cells)
- (vii) Usually smallest organelle in a cell.

Explain the effects of competitive and non-competitive inhibitors on the rate of enzymatic activity. [6]

Competitive

- competitive inhibitor / effect described in terms of competition ;
- complementary shape
- to active site **R** same / similar, structure to active site fits into active site ;
- blocking entry of substrate / prevents formation of ES complex ;
- lower rate at low substrate concentrations, but at high substrate concentration will reach the same plateau ;
- increasing substrate concentration reverses inhibition ;

Non-Competitive

- non-competitive inhibitor / described in terms of not competing ;
- fits into, a site other than active site / allosteric site ;
- shape of enzyme changes / shape of active site changes ;
- active site no longer complementary shape to substrate ;
- prevents formation of ES complex ;
- lower rate / no activity / does not reach the same rate at high substrate concentrations ;
- increase substrate concentration does not reverse inhibition ;

Explain the mode of action of enzymes in terms of specificity and activation energy. [10]

Enzymes are Specific

- Enzyme specificity depends on a close fit between substrate molecules and their binding sites on an enzyme
- Enzymes are highly specific due to the three dimensional configuration of its active site.
- This is described by the 'lock and key' hypothesis, where the substrate is imagined being like a key whose shape is complementary to the enzyme or lock.
- The active site is made up contact residues and the catalytic residues
- The catalytic residues are responsible for the ability of the enzyme to catalyse a particular metabolic reaction and they act on the bonds in the substrate(s).
- The contact residues are responsible for the specificity of the enzyme and form a shape that is complementary to the shape of the substrate.
- Molecules which do not fit nicely into the active site will not undergo a catalysed reaction
- When the correct substrate enters the active site, noncovalent or secondary forces bring substrate molecules into close contact with catalytic residues.
- Once the products are formed, the products no longer fit into the active site and escaped into the surrounding medium, leaving the active site free to receive other substrate.

Enzymes lower Activation Energy

1. Activation energy is the energy required to make the substances react.
2. The greater the activation energy, the slower the reaction at any particular temperature.
3. If the activation energy of a reaction decreased, the rate of reaction would be increased.
4. Enzymes hv functioning

Describe the effect of temperature on enzyme-catalyzed reactions. [4]

Temperature

- Heating increases molecular motion which facilitates a higher rate of successful collision between the enzymes and substrates.
- As a result, there is greater probability of a reaction occurring.
- A temperature increases, rate of enzyme-catalysed reactions increase.
- The temperature that promotes maximum activity is referred to as the optimum temperature.
- If the temperature is increased above this level, then a rapid decrease in the rate of the reaction occurs despite the increasing frequency of collisions.
- This is because the thermal agitation of the enzyme molecule disrupts the hydrogen bonds, ionic bonds, and hydrophobic interactions that stabilize the active conformation.
- The secondary and tertiary structures of the active site are gradually lost and the protein molecule eventually denatures.
- The denaturation of enzymes has a larger effect over the increase in molecular motion once the temperature rise past the optimum temperature.
- This accounts for the rapid decrease in reaction rate
- If temperature is reduced to near or below freezing point, enzymes are inactive, not denatured.